

MAT9004 Assignment 2

Due: Monday, 16 September 2024, 23:55 PM

Please write up your solutions neatly! In particular, use notation and terminology correctly and include some necessary comments. Partial marks will be given for showing that you know some aspects of the answer, even if your solution is incomplete. The questions are NOT arranged in order of difficulty, so you should attempt every question.

Both handwritten and typed solutions are accepted. Please make sure you convert all your work into **one single pdf file** as other types might not display properly on moodle.

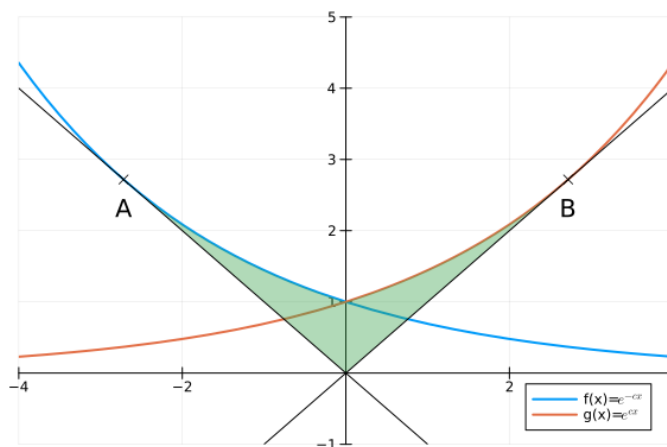


1. [10 points] An investment manager wants to invest \$1million dollars across multiple funds. Each investment fund has a different unit price and exposure to (or percentage of fund value invested in) shares, property and the bond market.

Fund	Unit Price	Shares	Property	Bonds	Return
A	\$3.25	20%	75%	5%	7%
B	\$2.50	35%	30%	35%	3%
C	\$6.75	45%	50%	5%	6%
D	\$4.00	75%	10%	15%	9%

The investment manager is required to have an average exposure of 50% to shares, 30% to property and 20% to bonds. Here exposure is again calculated as the percentage of the total dollar amount that is ultimately invested in each type of asset.

- [2 points] Let a, b, c, d be the amount in 1000's of dollars to be invested into each of the funds. Write a system of 4 linear equations to represent the requirement for the investment choice.
 - [2 points] Write these equations in matrix form. Explain, based on the application, why one of the equations might be redundant (not needed).
 - [4 points] Use Gaussian elimination to reduce the system of equations to upper triangular form and provide the solution to these equations.
 - [2 points] If return (% per dollar invested) for the last year, as given in the table above, is used as a guide for how profitable each fund is, how should the manager invest the money to maximise the return?
2. [8 points] Let $c > 0$ be some constant and consider the plots of the functions $f(x) = e^{-cx}$ and $g(x) = e^{cx}$ below. Let $A = (a, f(a))$ be the point where the tangent line of f goes through the origin and likewise $B = (b, g(b))$.



- [2 points] Find the equation of the tangent lines through A and B .
Hint: This requires solving for a, b but as a function of c .

- (b) [2 points] The tangent lines can be written in vector form as

$$\{\mathbf{v}_A t : t \in \mathbb{R}\} \text{ and } \{\mathbf{v}_B t : t \in \mathbb{R}\}$$

What are the vectors $\mathbf{v}_A$ and $\mathbf{v}_B$ and for what value of c are these orthogonal (at right angles)?

- (c) [4 points] What is the area of the shaded region bounded by the tangent lines and the functions $f(x)$ for $x \leq 0$ and $g(x)$ for $x \geq 0$? Provided the answer for arbitrary values of $c > 0$.

3. [12 points] Consider the matrix

$$A = \frac{1}{5} \begin{pmatrix} 39 & -48 \\ -48 & 11 \end{pmatrix}$$

- (a) [2 points] Show that if $\mathbf{v}$ is an eigenvector with eigenvalue λ of any matrix B , then $\mathbf{v}$ is also an eigenvector of the matrix (cB) for any constant $c \in \mathbb{R}$. What is corresponding eigenvalue of cB ?
- (b) [3 points] Hence or otherwise, state the characteristic polynomial and find the eigenvalues of A .
- (c) [3 points] Find the *set of all* eigenvectors for A corresponding to each of the eigenvalues.
- (d) [2 points] Show that the eigenvectors for the two different eigenvalues are orthogonal. Then select suitable eigenvectors so that the matrix $P = (\mathbf{v} \mid \mathbf{w})$ (i.e. the matrix formed by two eigenvectors in columns of V) has inverse $V^{-1} = V^T = \begin{pmatrix} \mathbf{v}^T \\ \mathbf{w}^T \end{pmatrix}$ (i.e. inverse formed by the same two eigenvectors in rows of V^T)
- (e) [2 points] For any $n \times n$ matrix B , if $B = PDP^{-1}$ with all diagonal entries D_{ii} of D non-zero, show that $B^{-1} = PD^{-1}P^{-1}$ where D^{-1} is a diagonal matrix with diagonal entries $\frac{1}{D_{ii}}$. Hence find A^{-1} as the product of 3 matrices (no need to multiply out the matrices)
4. [12 points] A new infectious disease has arrived in Australia. The effects of the disease are mild, but it is highly infectious. Epidemiologists have developed a model for the spread of the disease. They classify the population in 3 groups: u_t the number of people that have never been infected by day t , s_t the number of people sick on day t , and r_t the number of people that have recovered from one (or more) previous bouts of the disease by day t . They estimate that each day 20% of uninfected people fall sick, as well as 10% of those that have recovered from a previous bout. Each day 70% of those currently sick (s_t), recover. The disease arrives on day 0.

- (a) [3 points] Find a matrix A satisfying $\begin{pmatrix} u_{t+1} \\ s_{t+1} \\ r_{t+1} \end{pmatrix} = A \begin{pmatrix} u_t \\ s_t \\ r_t \end{pmatrix}$ for all $t \geq 0$. Then compute (u_2, s_2, r_2) if $u_0 = 25$ (millions of people) on the day when the disease first arrives in Australia.
- (b) [2 points] Show that $\begin{pmatrix} 6t \\ t \\ -7t \end{pmatrix}$ is an eigenvector of A for any t . What is the corresponding eigenvalue?
- (c) [2 points] Show that if $\mathbf{v} = (v_1, v_2)$ is an eigenvector of the matrix $C = \begin{pmatrix} c_{11} & c_{12} \\ c_{21} & c_{22} \end{pmatrix}$ then $\bar{\mathbf{v}} = (0, v_1, v_2)$ is an eigenvector of any matrix B of the form:

$$B = \begin{pmatrix} b_1 & 0 & 0 \\ b_2 & c_{11} & c_{12} \\ b_3 & c_{21} & c_{22} \end{pmatrix}$$

- (d) [3 points] Find the remaining two eigenvalues and an eigenvector corresponding to the largest eigenvalue
- (e) [2 points] Approximately what would the expected number of uninfected, sick & recovered people be after one year (365 days)?

Hint: Use the fact that with 3 distinct eigenvectors, $A = PDP^{-1}$ for suitable P, D and consider A^{365}

5. [8 points] Consider the set

$$S = \{(x, y) \in \mathbb{R}^2 : (x - 14)^2 + y^2 \leq 100, x^2 + (y - 14)^2 \leq 100, x + y \geq 14\} = T \cap U \cap P$$

where $T = \{(x, y) \in \mathbb{R}^2 : (x - 14)^2 + y^2 \leq 100\}$, $U = \{(x, y) \in \mathbb{R}^2 : x^2 + (y - 14)^2 \leq 100\}$, and $P = \{(x, y) \in \mathbb{R}^2 : x + y \geq 14\}$

- (a) [2 points] Describe very briefly what the sets T , U , P look like and the intersections $T \cap P$ and $U \cap P$
- (b) [4 points] Determine the points (if any) where the boundaries of each of the pairs of sets $T \& P$, $U \& P$ and $T \& U$ intersect, then draw the set S neatly.
- (c) [2 points] Hence or otherwise find the point(s) that minimise the function $f(x, y) = y$ over the set S .

Total: 50 points